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General Gate Teleportation and the Inner Structure of Its Clifford Hierarchies

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ABSTRACT

The quantum gate teleportation mechanism allows for the fault-tolerant implementation of "Clifford hierarchies" of gates assuming, among other things, a fault-tolerant implementation of the Pauli gates. We discuss how this method can be extended to assume the fault-tolerant implementation of any orthogonal unitary basis of operators, in such a way that each of these bases can be used to implement its own hierarchy of gates. Noticing that all of these hierarchies are defined through the same algebraic construction, we study this construction from a group-theoretic perspective. This allows us to deduce general results concerning all the hierarchies. We also discuss alternative hierarchies, study their properties, and compute some of their first few levels.

1 | Introduction

It is widely accepted that, at least in the near or medium term, the construction of reliable quantum computers will require the use of protocols for fault tolerance [1]. The *quantum gate teleportation* mechanism introduced by Gottesman and Chuang [2] enables the fault-tolerant implementation of an infinite hierarchy of gates, known as the *Clifford hierarchy*, which includes a universal set of quantum gates (Clifford + T) in its third level. The protocol requires three times as many physical qubits as logical qubits, and it assumes the ability to perform measurements in the Bell basis, the availability of some magic states, and a fault-tolerant implementation of the Pauli gates [3, 4]. The study of the Clifford hierarchy and its properties has been an active area of research in recent years [5–7], and, among other things, recent works [8] have studied its properties in qudit systems, and algorithms have been proposed to enumerate all the gates present in the different levels of the hierarchy.

In this work, we approach the study of gate teleportation and the Clifford hierarchy from a more abstract point of view, with the twofold aim of exploring all the possible use cases of the gate teleportation protocol and gaining new insights about the inner structure of the Clifford hierarchy.

In order to examine the full potential of gate teleportation, we devote Section 2 to discussing a more general approach to the protocol. We show how gate teleportation can be extended to any orthogonal unitary basis of operators—thus removing its dependence on the Pauli gates—in such a way that each of these

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bases defines its own gate teleportation mechanism, enabling the fault-tolerant implementation of its own "Clifford hierarchy" of gates. The practical advantage of considering this general protocol is that, under assumptions similar to those of standard gate teleportation, different choices of orthogonal unitary bases could lead to the fault-tolerant implementation of new gates, or it could make it easier to implement certain gates.

A close observation of Clifford hierarchies reveals that they all rely on the same algebraic construction. To our knowledge, this construction—which we have named as *quasinormalizer*—has not yet been studied from a group-theoretic point of view. We conduct such a study in Section 3, deducing some general properties about quasinormalizers that we then apply to Clifford hierarchies in Section 4. In particular, thanks to our formal framework, we are able to easily deduce that a bounded number of applications of any gate teleportation mechanism can never fault tolerantly implement an infinite amount of quantum gates from a finite set of fault-tolerant gates. We also show that the levels of the Clifford hierarchies can be understood as cosets of certain groups. This allows us to deduce general results about the cardinality of the levels and to more compactly represent all the elements of each level. In addition, we illustrate how our general formalism could help guide efforts in proving or disproving conjectures about Clifford hierarchies [8, Conj. 3.13].

Our general approach to quantum gate teleportation allows us to introduce alternative Clifford hierarchies of fault-tolerant gates, and we study some of these in Section 5, discussing properties such as their ability to implement T gates and drawing some connections with the exact synthesis of gates. In addition, in Appendix A, we extend the algorithm introduced by de Silva [8] to our general teleportation mechanism; we use this algorithm to symbolically compute the first few levels of some one-qubit hierarchies.

We conclude this work, in Section 6, by summarizing our results and discussing future work.

1.1 | Definitions and Conventions

In pursuit of generality and simplicity, we introduce the following definitions. Let $N > 1$ be a natural number. An N-dimensional (quantum) system is any quantum system having $\mathbb{C}^N$, or any isomorphic space, as its state space.

If $N = p_1^{n_1} \cdots p_l^{n_l}$ is the unique prime factorization of N, then an N-dimensional system is equivalent to a system with n_k p_k-dimensional qudits for $k = 1, \ldots, l$. For instance, if $N = 2^2 \cdot 3$, an N-dimensional system would be equivalent to a circuit with two qubits and one three-dimensional qudit.

For any natural number N, we denote the identity operator of $\mathbb{C}^N$ as I_N, and we let P_N be a set of *Pauli gates*: a set of representatives of the Pauli group for N-dimensional quantum systems in $U(N)/U(1)$ [9]. In this way, for example, we could have $P_2 = \{X, Y, Z, I_2\}$,

$$P_3 = \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{2\pi i/3} & 0 \\ 0 & 0 & e^{4\pi i/3} \end{pmatrix}^r \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}^s \;\middle|\; r, s \in \{0,1,2\} \right\},$$

and $P_{2^2 \cdot 3} = \{A \otimes B \otimes C | A, B \in P_2,\ C \in P_3\}$. We also define

$$|\Phi_N\rangle := \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle \otimes |k\rangle \in \mathbb{C}^N \otimes \mathbb{C}^N \simeq \mathbb{C}^{N^2},$$

as the Bell state on a pair of N-dimensional systems.

Given any unitary operation $V \in U(N)$ or any set $S \subseteq U(N)$ of unitary operations, we will use $[V]$ and $[S]$ to denote their respective projections onto $[U(N)] := U(N)/U(1)$.

For any group G (in particular, this would apply to the unitary group) with $x \in G$, we will use c_x to denote the function that maps any $g \in G$ to xgx^{-1}.

2 | General Gate Teleportation

Consider a set S of k-qubit gates containing all the Pauli gates on k qubits (the gates in P_{2^k}). If S can be implemented fault tolerantly, the quantum gate teleportation protocol introduced by Gottesman and Chuang [2] enables the fault-tolerant implementation of any gate $V \in U(2^k)$ such that $V P_{2^k} V^{-1} \subseteq S$, subject to the preparation of a magic state. This leads to the introduction of the k-qubit *Clifford hierarchy* as the sequence $(C_r)_{r \in \mathbb{N}}$, of sets of k-qubit gates up to phase, defined as $C_1 := [P_{2^k}]$ and

$$C_{r+1} := \left\{ V \in [U(2^k)] \;\middle|\; V[P_{2^k}]V^{-1} \subseteq C_r \right\},$$

for all $r \geq 1$. This scheme has been generalized to qudit systems [8, 9].

A natural question to ask upon looking at this gate teleportation mechanism is whether it could be further generalized beyond its dependence on Pauli gates. The answer to this question is affirmative and is the main focus of this section.

The algebra $\mathrm{End}(\mathbb{C}^N)$ of linear operators of $\mathbb{C}^N$ can be equipped with the Hilbert–Schmidt inner product, which maps any two operators A and B to the scalar $\mathrm{tr}(A^\dagger B)$, where $\dagger$ denotes Hermitian conjugation. With respect to this inner product, one may consider orthogonal bases of the algebra consisting, exclusively, of unitary operators; we will henceforth refer to such bases as *orthogonal unitary bases (of operators)*. As a notable example, the Pauli gates P_N form an orthogonal unitary basis of operators on $\mathbb{C}^N$. These bases have been extensively studied in the literature, with several works focusing on the more particular notion of *nice error bases* [10, 11].

Orthogonal unitary bases of operators are relevant to our discussion, for there is a known correspondence between them and quantum teleportation schemes [12]. In this section, we will show that each of these bases induces a gate teleportation

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mechanism—each defining a hierarchy analogous to the Clifford hierarchy. In this general scheme, the Gottesman–Chuang teleportation mechanism will be the one induced by the Pauli operators. Nevertheless, before we are able to properly discuss this, we need to cover a few preliminary results and definitions.

Let A be a linear operator on $\mathbb{C}^N$. In what follows, if $\{|r\rangle\}_{r=1}^N$ is the computational (or "canonical") basis of $\mathbb{C}^N$ and $A = \sum_{r,s} a_{rs}|r\rangle\langle s|$, we fix

$$A' \coloneqq \sum_{r,s} a_{rs}|s\rangle\langle r|, \qquad \bar{A} \coloneqq \sum_{r,s} \bar{a}_{rs}|r\rangle\langle s|,$$

to be the transpose and the complex conjugate of A (with respect to the computational basis). Notice that $\bar{a}_{kl}$ denotes the complex conjugate of a_{kl}.

Given any orthogonal unitary basis of operators B, it is straightforward to verify that B' and $\overline{B}$ are also orthogonal unitary bases of operators. Of course, B' and $\overline{B}$ refer to the sets containing, respectively, the transposes and complex conjugates of the elements of B (with respect to the computational basis).

Lemma 1. *If $B = \{B_r\}_{r=1}^{N^2}$ is an orthogonal unitary basis of operators on $\mathbb{C}^N$ and $|\Phi_N\rangle$ is the Bell state for a pair of N-dimensional quantum systems, then the family $\{(I_N \otimes B'_r)|\Phi_N\rangle\}_r$ is an orthonormal basis of $\mathbb{C}^{N^2}$.*

Proof. If $\{B_r\}_r$ is an orthogonal unitary basis, so is $\{B'_r\}_r$. It thus suffices to prove that $\{B_r|\Phi_N\rangle\}_r$ is an orthonormal basis of $\mathbb{C}^{N^2}$, which is known to be true [12, 4 ⇒ 3]. Nevertheless, for the sake of completeness, we include a simplified proof here. Given any pair $B_r \neq B_s$,

$$\begin{aligned}
&\langle\Phi_N|(I_N \otimes B_r)^\dagger(I_N \otimes B_s)|\Phi_N\rangle \\
&\quad = \langle\Phi_N|(I_N \otimes B_r^\dagger B_s)|\Phi_N\rangle \\
&\quad \propto \sum_{a,b}\langle a|b\rangle\langle a|B_r^\dagger B_s|b\rangle \\
&\quad = \sum_a \langle a|B_r^\dagger B_s|a\rangle = \operatorname{tr}(B_r^\dagger B_s) = 0;
\end{aligned}$$

hence, the vectors $(I_N \otimes B_r)|\Phi_N\rangle$ and $(I_N \otimes B_s)|\Phi_N\rangle$ are orthogonal. The cardinality of the set yields the desired result. □

Lemma 2. *Let A and V be unitary operators in $U(N)$, and let $|\alpha\rangle \in \mathbb{C}^N$ be the state of an N-dimensional system. Under these conditions,*

$$\begin{aligned}
&\Big(\big((I_N \otimes A)|\Phi_N\rangle\langle\Phi_N|(I_N \otimes A)^\dagger\big) \otimes I_N\Big) \\
&\big(|\alpha\rangle \otimes (I_N \otimes V)|\Phi_N\rangle\big) = |\phi\rangle \otimes (V\bar{A}|\alpha\rangle),
\end{aligned}$$

for some $|\phi\rangle \in \mathbb{C}^{2N}$.

Proof. If we expand some of the terms in our original expression, we get that

$$\begin{aligned}
(I_N \otimes A)|\Phi_N\rangle &= \tfrac{1}{\sqrt{N}}\sum_{r=0}^{N-1}|r\rangle \otimes A|r\rangle, \\
\langle\Phi_N|(I_N \otimes A)^\dagger &= \tfrac{1}{\sqrt{N}}\sum_{s=0}^{N-1}\langle s| \otimes \langle s|A^\dagger, \\
(I_N \otimes V)|\Phi_N\rangle &= \tfrac{1}{\sqrt{N}}\sum_{t=0}^{N-1}|t\rangle \otimes V|t\rangle.
\end{aligned}$$

Putting everything together, this leads us to

$$\begin{aligned}
&\Big(\big((I_N \otimes A)|\Phi_N\rangle\langle\Phi_N|(I_N \otimes A)^\dagger\big) \otimes I_N\Big)\big(|\alpha\rangle \otimes (I_N \otimes V)|\Phi_N\rangle\big) \\
&= \frac{1}{\sqrt{N^3}}\sum_{r,s,t}\big((|r\rangle\langle s|) \otimes \big(A|r\rangle\langle s|A^\dagger\big) \otimes I_N\big)(|\alpha\rangle \otimes |t\rangle \otimes V|t\rangle) \\
&= \frac{1}{\sqrt{N^3}}\sum_{r,s,t}(|r\rangle \otimes A|r\rangle) \otimes V|t\rangle \cdot \langle s|\alpha\rangle\langle s|A^\dagger|t\rangle \\
&= |\phi\rangle \otimes \sum_{s,t} V\langle s|\alpha\rangle\langle t|\bar{A}|s\rangle|t\rangle = |\phi\rangle \otimes \sum_s V\bar{A}\langle s|\alpha\rangle|s\rangle \\
&= |\phi\rangle \otimes V\bar{A}|\alpha\rangle,
\end{aligned}$$

where we have implicitly defined the ket:

$$|\phi\rangle = \frac{1}{\sqrt{N^3}}\sum_r |r\rangle \otimes A|r\rangle.$$

This concludes the proof. □

With these theoretical results, we are now prepared to introduce the general gate teleportation protocol. Consider an orthogonal unitary basis B of operators on $\mathbb{C}^N$ and a collection S (including B) of fault-tolerant N-dimensional quantum gates. We seek to fault tolerantly implement a gate V such that $VBV^{-1} \subseteq S$; physically, this condition means that we must be able to implement the conjugation VBV^{-1} fault tolerantly. For this purpose, we need a $3N$-dimensional quantum system, which we will subdivide into three N-dimensional subsystems. In this setup, the input to our gate V will be the state $|\alpha\rangle$ of the first subsystem, while the two other subsystems should be initialized to the magic state $(I_N \otimes V)|\Phi_N\rangle$. This is all depicted in Figure 1. Incidentally, this magic state is different from the kind of magic states considered in other protocols [13].

Assume that, under these conditions, and as shown in Figure 1, we perform a measurement in the basis $(I_N \otimes B')|\Phi_N\rangle$ on the first two subsystems (cf. Lemma 1). The classical outcome x of this measurement will uniquely determine a gate $A_x \in B$ such that the state of the whole system right after the measurement will be

$$\Big(\big((I_N \otimes A'_x)|\Phi_N\rangle\langle\Phi_N|(I_N \otimes A'_x)^\dagger\big) \otimes I_N\Big)\big(|\alpha\rangle \otimes (I_N \otimes V)|\Phi_N\rangle\big).$$

By virtue of Lemma 2, we know that the last N qubits will be in the state $V\bar{A}'_x|\alpha\rangle = VA_x^{-1}|\alpha\rangle$. Thus, if we apply the gate VA_xV^{-1} on the last N qubits, as shown in Figure 1, the resulting state will be

$$\big(VA_xV^{-1}\big)VA_x^{-1}|\alpha\rangle = V|\alpha\rangle,$$

thus allowing us to reach the desired state. Because, according to our hypotheses, the gate VA_xV^{-1} belongs to S, it can be implemented fault tolerantly. This procedure thus yields a fault-tolerant implementation of the gate V.

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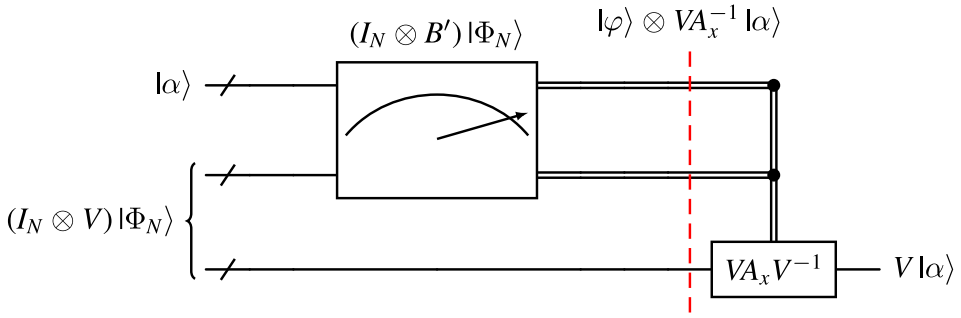


FIGURE 1 | Quantum circuit for the general teleportation scheme. This circuit aims to implement an n-dimensional gate V for some input $|\alpha\rangle$. [Colour figure can be viewed at wileyonlinelibrary.com]

This is how the generalized quantum gate teleportation protocol works. We summarize its capabilities in the following statement.

Claim 1. Let B be an orthogonal unitary basis of operators of $\mathbb{C}^N$, and assume that it possible to perform measurements in the $\{(I_N \otimes B'_r)|\Phi_N\rangle\}_r$ basis. Given any collection $S \supseteq B$ of fault-tolerant gates, any gate $V \in U(N)$ such that $VBV^{-1} \subseteq S$ can be implemented fault tolerantly as long as it is possible to prepare the state $(I_N \otimes V)|\Phi_N\rangle$.

We should highlight that when B is the orthogonal unitary basis P_N of Pauli gates, we retrieve the usual gate teleportation mechanism [2].

For an N-dimensional quantum system, given any orthogonal unitary basis B and a set of fault-tolerant gates $S \supseteq B$, the set of gates (up to phase) that can be implemented with the gate teleportation scheme induced by B is the set

$$\left\{ V \in [U(N)] \mid V[B]V^{-1} \subseteq [S] \right\} \qquad (\star)$$

Therefore, we define the *Clifford hierarchy generated by B on S* as the sequence of sets $C(B,S)$ with $C(B,S)_1 := [S]$ and, inductively,

$$C(B,S)_{r+1} := \left\{ V \in [U(N)] \mid V[B]V^{-1} \subseteq C(B)_r \right\},$$

for all $r \geq 1$. In this way, the gates in the rth level of the Clifford hierarchy generated by a basis V on a set S are the gates that can be implemented with up to $r-1$ applications of the gate teleportation protocol induced by B (provided that the gates in S are the only fault-tolerant gates available to us before being able to apply the protocol).

In order to realize any hierarchy $C(B,S)$, we must be able to fault tolerantly implement all the gates in B; hence, it makes sense to require $B \subseteq S$. Moreover, if all the gates in B can be implemented fault tolerantly, so can those of $\langle B \rangle$, the group generated by B, so it is natural to choose $S = \langle B \rangle$. This motivates the following definition.

Definition 1. Given any orthonormal unitary basis of operators B, we define the Clifford hierarchy generated by B to be

$$C(B) := C(B, \langle B \rangle).$$

We also define the *pure* Clifford hierarchy generated by B as

$$C_*(B) := C(B,B).$$

In principle, for any orthogonal unitary basis B, one can always consider the hierarchy $C(B,S)$ for any $B \subseteq S$. In regard to this general construction, we will just mention that,[1] given any such S and any natural numbers k and r,

$$C\big(B, C(B,S)_k\big)_r = C(B,S)_{k+(r-1)}.$$

In Section 5, we will study a few sample hierarchies. Nevertheless, for now, we can already notice that there are some obvious properties common to all Clifford hierarchies. For example, it is easy to see that all of them are increasing sequences of sets from the second level onwards. In the following section, we will take a more abstract and general approach that will later enable us to deduce some general results about these hierarchies.

3 | Quasinormalizers

Constructions of the form ($\star$) determine which gates can be implemented through any gate teleportation protocol in any general scenario, and in particular, they are used to recursively define all the different Clifford hierarchies induced by any orthogonal unitary basis. In this section, we will try to understand these constructions from a group-theoretical point of view. To this end, we introduce the following definition.

Definition 2. Let G be a group with two subsets $B, S \subseteq G$. The *quasinormalizer* of B in G with respect to S is the set

$$N_G(B,S) := \left\{ g \in G \mid gBg^{-1} \subseteq S \right\}.$$

The way quasinormalizers are related to our discussion of quantum gate teleportation is fairly direct.

Example 1. If B is an orthogonal unitary basis of operators on $\mathbb{C}^N$ and $B \subseteq S \subseteq U(N)$, the set of gates up to phase that can be implemented from S using the gate teleportation mechanism induced by B is $N_{[U(N)]}([B],[S])$. In this way, the Clifford hierarchy generated by B on S can be defined through $C(B,S)_1 := [B]$ and $C(B,S)_{r+1} = N_{[U(N)]}([B], C(B,S)_r)$.

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Through the remainder of this section, we will introduce some results that will help us understand the inner structure of quasinormalizers. The particularization to Clifford hierarchies of some of these results can be found in Proposition 3.

As we stated in the introduction, given an element x of a group G, we write c_x to denote the map $g \in G \mapsto xgx^{-1}$; clearly, if x and y belong to the same group, then $c_{xy} = c_x \circ c_y$. Moreover, in what follows, we will use $\mathrm{Cl}_G(B, S)$ to denote the set of conjugates of B that are included in S, that is,

$$\mathrm{Cl}_G(B, S) \coloneqq \{E \subseteq S | E = c_g(B) \text{ for some } g \in G\}.$$

We recall that, given a subset S of a group G, its *normalizer* $N_G(S)$ is the set of elements $g \in G$ such that $c_g(S) = S$ (the properties of this construction can be found in a book by Dummit and Foote [14]). Our choice of name and notation for quasinormalizers was motivated by the similarities between their definition and that of normalizers. In fact, as the following result will reveal, these two constructions can be equal under certain conditions.

Proposition 1. *Consider a group G with $B, S \subseteq G$.*

i. If $S \subseteq T \subseteq G$, then $N_G(B, S) \subseteq N_G(B, T)$.

ii. The normalizer of a finite subset is equal to its quasinormalizer with respect to itself: If B is a finite subset of a group G, then

$$N_G(B) = \{g \in G | c_g(B) = B\} = \{g \in G | c_g(B) \subseteq B\} = N_G(B, B).$$

iii. Given any subset $B \subseteq G$ and any subset $S \subseteq G$,

$$N_G(B, S) = N_G\big(B, \cup \mathrm{Cl}_G(B, S)\big).$$

This simple proposition should have awakened our intuition to quasinormalizers. We omit its proof as it is immediate, but a few remarks about it are in order. We should highlight that, for the purposes of (ii), there is no need for G to be finite. Regarding (iii), the result is showing us that, for the purposes of computing quasinormalizers, rather than looking at S, we should be looking at the conjugates of B that it includes.

Quasinormalizers may be groups—as the fact that they may be equal to normalizers clearly shows—but they do not have to be. A simple example can be found in the usual Clifford hierarchy: All its levels can be computed as quasinormalizers, as we have already discussed, and all the levels except the first two are known not to be groups [5]; incidentally, we will later provide a very succinct proof of this fact.

The fact that quasinormalizers are not necessarily groups makes their characterization immensely difficult from an algebraic point of view. In spite of this, we can still deduce a few interesting results that reveal some of the "structure" of quasinormalizers. We will discuss these results now.

Proposition 2. *Consider a group G with a pair of subsets B and S.*

i. The quasinormalizer $N_G(B, S)$ is a (disjoint) union of left cosets of $N_G(B)$.

ii. The quasinormalizer $N_G(B, S)$ is a (disjoint) union of right cosets of $N_G(S)$.

Proof. For (i), we will prove, by double inclusion, that

$$\bigcup_{x \in N_G(B,S)} xN_G(B) = N_G(B, S).$$

The inclusion from right to left is obvious because the identity element of G is included in $N_G(B)$, so we only need to take care of the inclusion from left to right.

Take an arbitrary element $x \in N_G(B, S)$. If $n \in N_G(B)$, then

$$c_{xn}(B) = c_x(c_n(B)) = c_x(B) \subseteq S,$$

so $xn \in N_G(B, S)$ for any $n \in N_G(B)$, which means that $xN_G(B) \subseteq N_G(B, S)$. As this holds for an arbitrary x in the quasinormalizer, our result follows.

The proof of (ii) is analogous to that of (i). Given any $x \in N_G(B, S)$, and any $n \in N_G(S)$, we have $c_{nx}(B) \subseteq c_n(S) = S$; hence, nx must belong to $N_G(B, S)$. □

Knowing that the quasinormalizers of sets B can be constructed as the (disjoint) unions of left cosets of the normalizer of B might make us wonder which cosets actually make up those unions. The following result sheds light on this question.

Theorem 1. *Let G be a group with $B, S \subseteq G$. If $\mathrm{Cl}_G(B, S) = \{c_{g_i}(B)\}_{i \in I}$ and $c_{g_i}(B) \neq c_{g_j}(B)$ for $i \neq j$, then the collection $\{g_i N_G(B)\}_{i \in I}$ is a partition of $N_G(B, S)$.*

Proof. Let us first prove, by double inclusion, that

$$N_G(B, S) = \bigcup_{i \in I} g_i N_G(B).$$

If $c_{g_i}(B) \in \mathrm{Cl}_G(B, S)$, then, clearly, $g_i \in N_G(B, S)$, so the inclusion from right to left follows from Proposition 2(i). For the remaining inclusion, consider an element $x \in N_G(B, S)$. Clearly, there has to exist an index $i \in I$ such that $c_x(B) = c_{g_i}(B)$; hence, $g_i^{-1}x \in N_G(B)$. Therefore, $x \in g_i N_G(B)$, as we wanted to show.

Lastly, let us show that the collection $\{g_i N_G(B)\}_i$ is pairwise disjoint. Given any two indices $i \neq j$, we know that $c_{g_i}(B) \neq c_{g_j}(B)$. This implies that $g_j^{-1} g_i \notin N_G(B)$ and, consequently, that $g_i \notin g_j N_G(B)$. Because cosets are either equal or disjoint, this concludes our proof. □

With these results, we can easily deduce some key properties about the cardinality of quasinormalizers.

Corollary 1. *Given a group G, consider two subsets $B, S \subseteq G$.*

i. The cardinality of $N_G(B, S)$ must be a multiple of the order of $N_G(S)$.

ii. We have $|N_G(B, S)| = |N_G(B)| \cdot |\mathrm{Cl}_G(B, S)|$.

iii. If $N_G(B)$ and S are finite, so must be $N_G(B, S)$.

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Proof. Statement (i) follows directly from Proposition 2. Statement (ii) can be deduced from Theorem 1, and Statement (iii) follows from it. □

In the following section, we will show how these results can be applied to obtain general properties about Clifford hierarchies.

Remark 1. Consider a group G with $B, S \subseteq G$. The notion of a "quasinormalizer" could more accurately be labeled as that of a left quasinormalizer, for it would be possible to define the dual construction of a right quasinormalizer as

$$N_G^{\rightarrow}(B,S) = \{g \in G | g^{-1}Bg \subseteq S\}.$$

It can be easily verified that

$$N_G^{\rightarrow}(B,S) = \left(N_G(B,S)\right)^{-1}.$$

It thus easily follows that—just as left quasinormalizers $N_G(B,S)$ are unions of *left* cosets of $N_G(B)$—right quasinormalizers $N_G^{\rightarrow}(B,S)$ are unions of *right* cosets of $N_G(B)$. Analogously, any structural property of left quasinormalizers has its corresponding dual property for right quasinormalizers.

4 | Consequences for Clifford Hierarchies

Now, that we have an abstract understanding of the construction behind the definition of all Clifford hierarchies, we can use our results to deduce some general properties about them.

We shall first discuss some obvious consequences of Propositions 1 and 2 and Corollary 1, which are a mere particularization of these results.

Proposition 3. *For any natural N, consider an orthogonal unitary basis B of operators acting on $\mathbb{C}^N$, together with a set $B \subseteq S \subseteq U(N)$. Let $C \coloneqq C(B,S)$ be the Clifford hierarchy generated by B on S. Consider any natural $r \geq 2$.*

i. The level C_r is a (disjoint) union of left cosets of $C_*(B)_2 = N_{[U(N)]}(B)$ and a (disjoint) union of right cosets of $N_{[U(N)]}(C_{r-1})$.

ii. The cardinality of C_r must be divisible by the order of $C_*(B)_2 = N_{[U(N)]}$ and by the order of the normalizer of C_{r-1} in $[U(N)]$.

In particular, Statement (i) in the proposition above allows for a very compact representation of all the levels of Clifford hierarchies. We will take advantage of this in Section 5.

It is a known fact [6] that all the levels of the usual Clifford hierarchy are finite. We now seek to prove an analogous statement for all possible Clifford hierarchies, generated by any orthogonal unitary basis of operators. To that end, we need the following result.

Lemma 3. *Given a natural N, let $E \subseteq [U(N)]$ be a finite set of unitary operators up to phase. If E contains the projection of a basis of $\mathrm{End}(\mathbb{C}^N)$, then the order of its normalizer in $[U(N)]$ is finite.*[2]

Proof. Over the course of this proof, given any $P \in [U(N)]$, we will use $\overline{P}$ to denote any unitary operator such that $[\overline{P}] = P$. We shall extend this notation, in the obvious way, to subsets of $[U(N)]$.

Let B be a basis of the vector space of endomorphisms of $\mathbb{C}^N$ such that $[B] \subseteq E$. Given any operator X on $\mathbb{C}^N$, we shall consider the set $r(X)$ of all operators $Y \in [X]$ whose characteristic polynomial is equal to the characteristic polynomial of some operator in B. In other words,

$$r(X) \coloneqq \{Y \in [X] | p(Y) = p(B_i) \text{ for some } B_i \in B\},$$

where we use $p(-)$ to denote the characteristic polynomial of an operator. Because B is finite, this collection must be finite[3] for any operator X. In addition to this, we will consider the set

$$R \coloneqq \bigcup_{X \in \bar{E}} r(X),$$

which, because $\bar{E}$ is finite, must be finite itself.

Let L be the set of all linear operators Λ on the vector space $\mathrm{End}(\mathbb{C}^N)$ such that $\Lambda(B) \subseteq R$; notice that Λ maps linear operators to linear operators. Formally,

$$L \coloneqq \{\Lambda \in \mathrm{End}(\mathrm{End}(\mathbb{C}^N)) | \Lambda(B) \subseteq R\}.$$

Because B is a basis of $\mathrm{End}(\mathbb{C}^N)$, these operators Λ are fully determined by how they map the elements of B. Therefore, as both B and R are finite, so has to be L.

We will now prove that the normalizer of E in $[U(N)]$ can be embedded into L and must therefore be finite. To this end, it suffices to notice that the function

$$\begin{aligned} N_{[U(N)]}(E) &\to L \\ P &\mapsto c_{\overline{P}}, \end{aligned}$$

which takes any element to the conjugation action of any of its representatives, is well defined and injective. Firstly, it is obvious that the choice of representative of P has no effect in the resulting conjugation action. In addition, the map does map into L. Indeed, consider some P in the normalizer of E and some $B_i \in B$. Clearly, $[c_{\overline{P}}(B_i)] \in [X]$ for some $X \in \bar{E}$. Moreover, because the characteristic polynomial of an operator is preserved under conjugation, $c_{\overline{P}}(B_i) \in r(X) \subseteq R$, just as we wanted to show. Lastly, the injectivity of this assignment can be deduced from the fact that the conjugation action of two operators is identical if and only if one is a scalar multiple of the other.

This shows that the normalizer of E can be embedded into L and, therefore, that it must be finite. □

Theorem 2. *For some natural number N, let B be an orthogonal unitary basis of linear operators on $\mathbb{C}^N$, and let $S \subseteq U(N)$ be a finite set of unitary operators including B. All the levels of the Clifford hierarchy $C(B,S)$ generated by B on S are finite. Moreover, for any natural $r > 1$,*

$$|C_r(B,S)| \leq |N_{[U(N)]}([B])|^{r-1} \cdot |\mathrm{Cl}([B],[S])|.$$

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Proof. The bound on the cardinality for $r = 2$ is a trivial consequence of Corollary 1(ii). The bound for an arbitrary r follows by induction, using the bound $|\mathrm{Cl}([B], C_{r-1})| \leq |C_{r-1}|$. Lastly, the finitude of all the levels of the hierarchy can be deduced from this bound taking into account that both S and the normalizer of $[B]$ are finite (cf. Lemma 3). □

From a more physical point of view, this theorem is simply stating that—if one only has access to a finite collection of fault-tolerant gates—a bounded number of applications of any gate teleportation schemes can only yield a finite amount of fault-tolerant gates. With our theoretical framework, we have been able to easily prove this result that generalizes a known property of the usual Clifford hierarchy.

As a consequence of Theorem 2, we may also deduce that no level of any Clifford hierarchy may be a group if it includes a set of universal quantum gates, for such a set would generate an infinite group. In particular, this shows why the levels of the usual Clifford hierarchy are not groups from the third level onwards.

In a recent work by de Silva, it is conjectured [8, Conj. 3.13] that "all conjugate tuples of k-th level gates are[4] if all the operators included in the group that they generate (in $[U(N)]$) belong to the k-th level of the Clifford hierarchy k-closed." In a more general language, this statement can be rephrased as the slightly stronger conjecture that if a level of the Clifford hierarchy includes a generating subset of a conjugate of the Pauli group, then it must contain the whole conjugate. While we have not solved the conjecture for the particular case of the Clifford hierarchy, we have found that it would not hold if considered for arbitrary groups.

Proposition 4. *Let G be a group with a subgroup $B \leq G$ and a subset $S \subseteq G$. Even if the quasinormalizer $N_G(B, S)$ includes a generating subset of a conjugate $c_g(B)$ of B, the conjugate itself may not be included in $N_G(B, S)$.*

Proof. Let G be the symmetric group of order eight with function composition $\circ$ as operation. We recall [14] that the symmetric group of order n contains the permutations of the set $\{1, \ldots, n\}$, that is, the bijective functions from $\{1, \ldots, n\}$ to itself. These permutations are represented using *cycle notation*, in such a way that, for example, $(1, 2, 3)(4, 5)$ is the permutation mapping $1 \mapsto 2$, $2 \mapsto 3$, $3 \mapsto 1$, $4 \mapsto 5$, and $5 \mapsto 4$, and leaving any other numbers unchanged. We should also mention that, given any permutation σ and any subset S of a symmetric group, σS represents the set of all permutations of the form σs with $s \in S$ (and, here, σs should be understood as $\sigma \circ s$, as we are using multiplicative notation).

If we take

$$C_1 \coloneqq B \coloneqq \langle (1, 2), (3, 4)(5, 6)(7, 8)(1, 2) \rangle,$$

where we use $\langle - \rangle$ to denote the group generated by a subset, then—with the help of a computer—it is easy to check that the sets $C_2 \coloneqq N_G(B, C_1)$, $C_3 \coloneqq N_G(B, C_2)$, and $C_4 \coloneqq N_G(B, C_4)$ are equal to the decompositions shown in Figure 2.

A simple (computer-assisted) verification reveals that C_4 contains a generating subset of the conjugate of B

$$\langle (8, 4), (3, 6)(5, 2)(7, 1)(8, 4) \rangle$$

$$\begin{aligned}
C_2 &\coloneqq N_G(B, C_1) = N_G(B), \\
C_3 &\coloneqq N_G(B, C_2) = C_2 \cup (6, 4)N_G(B) \cup \\
&\quad (8, 6)N_G(B) \cup (8, 4)N_G(B) \cup \\
&\quad (7, 1)(8, 2)N_G(B) \cup (7, 1)(8, 2)(6, 4)N_G(B) \cup \\
&\quad (7, 6, 8)N_G(B) \cup (7, 4, 8)N_G(B) \cup \\
&\quad (5, 1, 7)(8, 6, 2)N_G(B) \cup (5, 1, 7)(8, 4, 6, 2)N_G(B) \cup \\
&\quad (5, 1, 7, 4, 6, 2, 8)N_G(B) \cup (5, 4, 8, 6, 7)N_G(B) \cup \\
&\quad (5, 4, 8, 2, 6, 7, 1)N_G(B) \cup (3, 1, 7, 5)(8, 6, 4, 2)N_G(B) \cup \\
&\quad (3, 1, 7, 5)(8, 4, 2)N_G(B) \cup (3, 1, 7, 4, 2, 8, 5)N_G(B), \\
C_4 &\coloneqq N_G(B, C_3) = C_3 \cup (7, 1)(8, 6, 2)N_G(B) \cup (7, 1)(8, 6, 2, 4)N_G(B) \cup \\
&\quad (7, 1, 4, 8, 6, 2)N_G(B) \cup (7, 1)(8, 2, 6)N_G(B) \cup \\
&\quad (7, 1)(8, 2, 4)N_G(B) \cup (7, 1, 6)(8, 2)N_G(B) \cup \\
&\quad (7, 1, 4, 8, 2)N_G(B) \cup (7, 1)(8, 4, 2, 6)N_G(B) \cup \\
&\quad (7, 1)(8, 4, 2)N_G(B) \cup (7, 1, 6)(8, 4, 2)N_G(B) \cup \\
&\quad (7, 6, 2, 8, 1)N_G(B) \cup (7, 6, 2, 4, 8, 1)N_G(B) \cup \\
&\quad (7, 6, 2)(8, 1, 4)N_G(B) \cup (7, 4, 2, 6, 8, 1)N_G(B) \cup \\
&\quad (7, 4, 2, 8, 1)N_G(B) \cup (7, 4, 2, 8, 1, 6)N_G(B) \cup \\
&\quad (5, 1, 8, 7)(6, 2)N_G(B) \cup (5, 1, 8, 7)(6, 4, 2)N_G(B) \cup \\
&\quad (5, 1, 6, 2, 8, 7)N_G(B) \cup (5, 1, 6, 4, 2, 8, 7)N_G(B) \cup \\
&\quad (5, 1, 7)(8, 6, 2, 4)N_G(B) \cup (5, 1, 7)(8, 6, 4, 2)N_G(B) \cup \\
&\quad (5, 1, 4, 8, 6, 2, 7)N_G(B) \cup (5, 1, 7)(8, 2)N_G(B) \cup \\
&\quad (5, 1, 7)(8, 2, 4)N_G(B) \cup (5, 1, 4, 8, 2, 7)N_G(B) \cup \\
&\quad (5, 1, 7, 2, 8)N_G(B) \cup (5, 1, 7, 2, 4, 8)N_G(B) \cup \\
&\quad (5, 1, 4, 8)(7, 2)N_G(B) \cup (5, 4, 2, 6, 1, 8, 7)N_G(B) \cup \\
&\quad (5, 4, 2, 8, 7)(1, 6)N_G(B) \cup (5, 4, 2, 8, 6, 1, 7)N_G(B) \cup \\
&\quad (3, 1, 8, 7, 5)(6, 2)N_G(B) \cup (3, 1, 8, 7, 5)(6, 4, 2)N_G(B) \cup \\
&\quad (3, 1, 6, 2, 8, 7, 5)N_G(B) \cup (3, 1, 6, 4, 2, 8, 7, 5)N_G(B) \cup \\
&\quad (3, 1, 7, 5)(8, 6)(2, 4)N_G(B) \cup (3, 1, 7, 5)(8, 6, 2)N_G(B) \cup \\
&\quad (3, 1, 4, 2, 8, 6, 7, 5)N_G(B) \cup (3, 1, 7, 5)(8, 2, 4, 6)N_G(B) \cup \\
&\quad (3, 1, 7, 5)(8, 2)(6, 4)N_G(B) \cup (3, 1, 4, 6, 7, 5)(8, 2)N_G(B) \cup \\
&\quad (3, 1, 7, 2, 4, 6, 8, 5)N_G(B) \cup (3, 1, 7, 2, 8, 5)(6, 4)N_G(B) \cup \\
&\quad (3, 1, 4, 6, 7, 2, 8, 5)N_G(B) \cup (3, 1, 6, 7)(5, 2, 4, 8)N_G(B) \cup \\
&\quad (3, 1, 4, 8, 5, 2, 6, 7)N_G(B) \cup (3, 1, 5, 2, 6, 7)(8, 4)N_G(B).
\end{aligned}$$

FIGURE 2 | Representation of the sets C_2, C_3, and C_4 as (disjoint) unions of left cosets of $N_G(B)$ (proof of Proposition 4).

but that the conjugate itself is not included within C_4, hence providing us with a counterexample.

We performed this computations using GAP [15]. It is worth noting that the product in the symmetric group used by GAP is $f \cdot g \coloneqq g \circ f$. This should be kept in mind if trying to reproduce the results. □

In spite of how disconnected this result may appear to be from the original conjecture, it reveals something about it. The counterexample that we have provided in the proof means that either the conjecture is false or, if it is true, it cannot be deduced from the algebraic structure of the Clifford hierarchy as a nested sequence of quasinormalizers. Hence, if there is a proof of that conjecture for the Clifford hierarchy, it will necessarily have to rely on the intrinsic properties of the hierarchy, and—unlike the ones we have discussed in this section—it will not be deducible as the particular case of a structural property of quasinormalizers.

This shows how our formalism can be used to guide efforts to explore properties about the Clifford hierarchy. In an initial

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approach to a problem, it might be challenging to try to look for counterexamples within the Clifford hierarchy itself, but exploring analogous hierarchies in finite groups is much more feasible and can provide useful hints on how the problem might be addressed.

5 | Some Clifford Hierarchies

We shall now discuss some properties of the Clifford hierarchy and introduce new examples of Clifford hierarchies that can be constructed within our general framework. In each of these examples, we will consider an orthogonal unitary basis of operators in $\mathrm{End}(\mathbb{C}^2)$ and explore some of the properties of the Clifford hierarchies (pure and nonpure) that they generate.

In particular, for each hierarchy under consideration, we will comment on the minimum level that includes the T gate; we refer to this level as the *T-level* of the hierarchy. If the T gate does not belong to a hierarchy, we fix its T-level to be infinity.

The minimum level at which a gate is included in a hierarchy can be determined or lower bounded with a simple recursive procedure, which we now outline.

Algorithm 1. Given a hierarchy of the form $C(B, S)$ and a gate V:

1. If $[V] \in [S]$, the minimum level that includes V is 1. Otherwise, continue.
2. Compute the set VBV^{-1} and find the minimum levels in which these elements are included. Out of all of these levels, find the maximum m.
3. The minimum level at which V is included is $m + 1$.

Of course, this algorithm would run indefinitely if V does not belong to the hierarchy. In this case, the best that this procedure can give is a lower bound for the level, which can be obtained from the maximum recursion depth reached.

Regarding this notion of the T-level, the following result can be illuminating.

Proposition 5. *In a qubit system of any dimension (i.e., a quantum system of some dimension 2^n), if B is an orthogonal unitary basis that does not include the T gate, then the T-level of the pure hierarchy generated by B cannot be less than* 3.

Proof. Consider a quantum system of dimension 2^n, and let $m := 2^{n-1}$ so that $2^n = 2m$. Given any orthogonal unitary basis B not containing T gates, we will prove that the pure hierarchy generated by B cannot have T-level 2. This is equivalent to showing that, for any gate $V \in B$, we have $c_T(V) \notin B$, where T actually represents the one-qubit T gate tensored with identity gates.

Let $\omega := e^{i\pi/4}$. Without loss of generality, we will consider T to be a T gate on the first qubit:

$$T := \begin{pmatrix} 1 & 0 \\ 0 & \omega \end{pmatrix} \otimes I_m.$$

Given any $V \in B$ with coordinate matrix $(v_{jk})_{jk}$ with respect to the computational basis, it is easy to verify that the matrix entries of $c_T(V)$ are given by

$$c_T(V)_{jk} = \begin{cases} v_{jk} & \text{if} j, k \le m \text{ or } j, k > m, \\ \omega v_{jk} & \text{if } j < m \text{ and } k > m, \\ \overline{\omega} v_{jk} & \text{if } j < m \text{ and } k > m. \end{cases}$$

It then follows that the Hilbert–Schmidt inner product of V with $c_T(V)$ is

$$\mathrm{tr}\left(V^\dagger c_T(V)\right) = \sum_{\substack{j,k\le m,\\ k,j>m}} |v_{jk}|^2 + \omega\left(\sum_{k<m<j} |v_{jk}|^2\right) + \overline{\omega}\left(\sum_{j<m<k} |v_{jk}|^2\right),$$

which cannot be zero because $V \neq 0$ and all the nonzero addends carry a positive real contribution. Because $c_T(V)$ and V are therefore not orthogonal and B is an orthogonal unitary basis, we can conclude that $c_T(V) \notin B$. □

de Silva introduced an algorithm for enumerating the elements of any level of the usual Clifford hierarchy [8, §3] (generated by the Pauli group). In Appendix A, and building upon the ideas present in that work, we present an algorithm that enables us to compute the elements of any level of any general Clifford hierarchy $C(B, S)$, only under the assumption that S be finite. Through the current section, we use this algorithm to symbolically compute the first few levels of some of the one-qubit hierarchies that we discuss. We also rely on Proposition 3 in order to represent our results in a compact manner.

Remark 2. In an initial attempt, we tried to run the algorithm numerically, but the results appeared to be inconsistent. A closer examination revealed that the algorithm was very sensitive to numerical errors. Thus, we have implemented our algorithm using SymPy [16].

This has prevented us from computing complex hierarchies. A particular obstacle that we have faced is the fact that, a priori, there is no way of fixing a finite algebraic extension of the rational numbers in which to execute the algorithm.

5.1 | Pauli Basis

As we have already mentioned, the Clifford hierarchy $C(P_{2^n})$ generated by the n-qubit Pauli group is the standard n-qubit hierarchy. Moreover, because the Pauli basis forms a group in $[U(2^n)]$, which is isomorphic to the direct product of n copies of the Klein four-group, we have $C(P_{2^n}) = C_*(P_{2^n})$. The T-level of these hierarchies is 3.

We now present a result that links the Clifford hierarchy with the exact synthesis of quantum gates.

Theorem 3. *Let n and $k \ge 2$ be natural numbers. Using ancilliary qubits, any matrix $V \in U(2^n)$ with entries in*

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$\mathbb{Z}[1/2, \exp(2\pi i/2^k)]$ *can be exactly represented using the gates in the* k*th level of the* n*-qubit Clifford hierarchy.*

Proof. Throughout this proof, we will denote $\omega_k := e^{2\pi i/2^k}$ and

$$P(x) := \begin{pmatrix} 1 & \\ & x \end{pmatrix}, \qquad Q := \begin{pmatrix} & 1 \\ y & \end{pmatrix}.$$

According to a recent work by Amy et al. [17], we know that—with access to enough ancillary qubits—in order to exactly represent the quantum gates with entries in $\mathbb{Z}[1/2, \omega_k]$, only the gates H and $P(\omega_k)$ are necessary, together with the CNOT gate if $n > 1$. It is well known that the H gate belongs to the Clifford group (the second level of the Clifford hierarchy) and that so does the CNOT gate when $n > 1$; hence, we only need to take care of the $P(\omega_k)$ gate. For that, it will suffice to show that the gate $T(\omega_k)$ belongs to the kth level of the one-qubit Clifford hierarchy C.

It is easy to check that, for any $y \in U(1)$, we have

$$[c_{Q(y)}(X)] = [Q(y^2)], \quad [c_{Q(y)}(Y)] = [Q(-y^2)], \quad [c_{Q(y)}(Z)] = [Z];$$

hence, if $[Q(\pm y^2)] \in C_k$, then $[Q(\pm y)] \in C_{k+1}$. Because $[Q(\omega_1)] = [Y]$ and $[Q(-\omega_1)] = [Y]$ both belong to C_1, we can conclude by induction that every level k contains the gates $[Q(\pm\omega_k)]$.

Following an analogous procedure, we can check how, for every $x \in U(1)$,

$$[c_{P(x)}(X)] = [Q(x^2)], \quad [c_{P(x)}(Y)] = [Q(-x^2)], \quad [c_{Q(x)}(Z)] = [Z],$$

so if $[Q(\pm x^2)] \in C_k$, then $[P(\pm y)] \in C_{k+1}$, and by the conclusion in the previous paragraph, it follows that $[P(\omega_k)] \in C_k$ for every $k > 1$ (the case $k = 1$ is trivial). This completes the proof. □

We should also mention that the composition of the standard Clifford hierarchy has been characterized for one-qudit systems of prime dimension [18]. This includes, in particular, one-qubit systems.

5.2 | Basis H

Given an orthogonal unitary basis B, if σ is an automorphism of the algebra of unitary operators, $\sigma(B)$ will also be an orthogonal unitary basis and its Clifford hierarchy on any set S will be algebraically isomorphic[5] to the one generated by B on $\sigma^{-1}(S)$. We now consider the image of the Pauli basis through an automorphism.

The orthogonal unitary basis

$$B_H := \left\{ \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, Y, I_2 \right\}$$

can be obtained as the image of an automorphism on the Pauli basis $\{X, Z, Y, I_2\}$. Consequently, $[B_H]$ also forms a group isomorphic to the Klein four-group, and we have $C(B_H) = C_*(B_H)$. Moreover, this one-qubit hierarchy is algebraically isomorphic to the usual hierarchy $C(P_2)$. Nevertheless, and in contrast to the standard Clifford hierarchy, the T-level of $C(B_H)$ is not 3; according to our computations, it is more than 16.

As a consequence of the above, obtaining the elements of the Basis H hierarchy from the standard Clifford hierarchy is direct. Indeed, if we let $\langle \cdot, \cdot \rangle$ denote the Hilbert–Schmidt inner product, the function

$$\sigma(V) := \frac{1}{\sqrt{2}}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\frac{\langle X, V\rangle}{\langle X, X\rangle} + H\frac{\langle Z, V\rangle}{\langle Z, Z\rangle} + Y\frac{\langle Y, V\rangle}{\langle Y, Y\rangle} + I_2\frac{\langle I_2, V\rangle}{\langle I_2, I_2\rangle}$$

defines an automorphism that maps the Pauli basis to Basis H. Therefore, for any natural k, the kth level of the Basis H hierarchy will just be the image of σ on the kth level of the usual Clifford hierarchy:

$$C(B_H)_k = \sigma\big(C(P_2)_k\big).$$

We will now introduce some hierarchies that are not isomorphic to the one generated by the Pauli basis.

5.3 | Basis S

The family

$$B_S := \left\{ \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix}, Y, X \right\}$$

forms an orthogonal unitary basis. Its projection onto $[U(2)]$ does not form a group, and the group that it generates is isomorphic to the dihedral group of order 8. Consequently, $C(B_S) \neq C_*(B_S)$. The pure hierarchy $C_*(B_S)$ has T-level 3, whereas $C(B_S)$ has T-level 2. This means that the T-level of $C_*(B_S)$ is equal to that of the standard Clifford hierarchy and optimal among pure hierarchies. In contrast, the T-level of $C(B_S)$ is even lower.

We first ran our algorithm to compute some of the levels of the pure hierarchy $C_*(B_S)$, but we found that its second level had 8 elements and included B_S. We know that the group generated by B_S also has 8 elements; hence, it must be equal to the second level of the pure hierarchy; this follows from the fact that $C_*(B_S)_2 = N_{[U(N)]}(B_S)$ is a group.

We can therefore conclude that, for any natural number k,

$$C(B_S)_k = C_*(B_S)_{k+1}.$$

Moreover, as per Proposition 3, all the levels of the hierarchy generated by B_S can be represented as left cosets of its first level because

$$N_{[U(N)]}(B) = C_*(B_S)_2 = \langle B\rangle = C(B_S)_1.$$

We have computed the first six levels of $C(B_S)$, and we present the cosets that make up these levels in Figure 3. From this figure, we can deduce that the cardinalities of Levels 1–6 are, respectively, 8, 16, 32, 64, 128, and 256. Based on the levels that we have computed, we observe and conjecture that the Basis S hierarchy exhibits a consistent and predictable composition, unlike the other hierarchies that we have studied.

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Level	New cosets of $C(B_S)_1$ in $C(B_S)$			
1	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$			
2	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$			
3	$\begin{pmatrix} 0 & e^{i\frac{\pi}{8}} \\ 1 & 0 \end{pmatrix}$		$\begin{pmatrix} 0 & e^{-i\frac{\pi}{8}} \\ 1 & 0 \end{pmatrix}$	
4	$\begin{pmatrix} 0 & e^{i\frac{\pi}{16}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{\pi}{16}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{3\pi}{16}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{5\pi}{16}} \\ 1 & 0 \end{pmatrix}$
5	$\begin{pmatrix} 0 & e^{i\frac{\pi}{32}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{i\frac{3\pi}{32}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{\pi}{32}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{3\pi}{32}} \\ 1 & 0 \end{pmatrix}$
	$\begin{pmatrix} 0 & e^{-i\frac{5\pi}{32}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{7\pi}{32}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{9\pi}{32}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{11\pi}{32}} \\ 1 & 0 \end{pmatrix}$
6	$\begin{pmatrix} 0 & e^{i\frac{\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{i\frac{3\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{i\frac{5\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{i\frac{7\pi}{64}} \\ 1 & 0 \end{pmatrix}$
	$\begin{pmatrix} 0 & e^{-i\frac{\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{3\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{5\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{7\pi}{64}} \\ 1 & 0 \end{pmatrix}$
	$\begin{pmatrix} 0 & e^{-i\frac{9\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{11\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{13\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{15\pi}{64}} \\ 1 & 0 \end{pmatrix}$
	$\begin{pmatrix} 0 & e^{-i\frac{17\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{19\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{21\pi}{64}} \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & e^{-i\frac{23\pi}{64}} \\ 1 & 0 \end{pmatrix}$

FIGURE 3 | The first six levels of the Clifford hierarchy generated by Basis S.

Unlike in the standard Clifford hierarchy, the first few levels of the Basis S hierarchy do not include a universal set of quantum gates. To fix this, we can assume the fault-tolerant implementation of a Hadamard gate and consider the *adulterated Basis S hierarchy* $C(B_S, \langle B_S, H\rangle)$. Because $\langle B_S, H\rangle$ is the Clifford group, the first level of this hierarchy will agree with the second level of the standard one-qubit hierarchy, but the rest of the levels will be different. To illustrate what new cosets are brought to the hierarchy by the addition of a Hadamard gate to its first level, these would be the additional cosets of $C(B_S)_1 = C_*(B_S)_2 = \langle B_S\rangle$ that would lie in the second level of $C(B_S, \langle B_S, H\rangle)$:

$$\frac{1}{\sqrt{2}}\begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}, \quad \frac{1}{\sqrt{2}}\begin{pmatrix} i & i \\ 1 & -1 \end{pmatrix}, \quad \frac{1}{2}\begin{pmatrix} -\sqrt{2} & 1+i \\ \sqrt{2} & 1+i \end{pmatrix},$$
$$\frac{1}{2}\begin{pmatrix} 2\sqrt{i} & 1-i \\ \sqrt{2} & 1+i \end{pmatrix}, \quad \begin{pmatrix} 0 & e^{i\frac{\pi}{8}} \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & e^{-i\frac{7\pi}{8}} \\ 1 & 0 \end{pmatrix}.$$

The Basis S hierarchy behaves in a similar way to the standard hierarchy in terms of our previous result on the exact synthesis of gates.

Proposition 6. *Let $k \geq 1$ be a natural number. Using ancilliary qubits, any matrix $V \in U(2)$ with entries in $\mathbb{Z}[1/2, \exp(2\pi i/2^k)]$ can be exactly represented using the gates in the $(k-1)$th level of the adulterated Basis S hierarchy $C(B_S, \langle B_S, H\rangle)$.*

Proof. The proof is perfectly analogous to that of Theorem 3, restricted to one qubit. It suffices to take into account that

$$[c_{P(x)}(S)] = [S], \quad [c_{P(x)}(S^\dagger)] = [S^\dagger], \quad [c_{Q(y)}(S)] = [S^\dagger], \quad [c_{Q(y)}(S)] = [S],$$

for any $x, y \in U(1)$. □

Notice that, with the Basis S hierarchy, we require the $(k-1)$th level instead of the kth level. If we extend the Basis S hierarchy to several qubits using tensor products, we can formulate an analogous result.

Proposition 7. *Let $n > 1$ and $k \geq 3$ be natural numbers and define the sets*

$$B_S^n = \{A_1 \otimes \cdots \otimes A_n | A_1, \ldots, A_n \in B_S\},$$
$$E_S^n := \langle A_1 \otimes \cdots \otimes A_n | A_1, \ldots, A_n \in B_S \cup \{H, I_2\}\rangle.$$

Using ancilliary qubits, any matrix $V \in U(2)$ with entries in $\mathbb{Z}[1/2, \exp(2\pi i/2^k)]$ can be exactly represented using the gates in the $(k-1)$th level of the n-qubit hierarchy $C(B_S^n, E_S^n)$.

Proof. By the construction of B_S and E_S, it can be readily checked that any one-qubit gate belonging to the kth level of the one-qubit hierarchy $C(B_S, \langle B_S, H\rangle)$ will also belong (taking the appropriate tensor products with the identity gate) to the kth level of $C(B_S^n, E_S^n)$. Similarly, any two-qubit gate in the kth level of $C(B_S^2, E_S^2)$ will belong to the kth level of $C(B_S^n, E_S^n)$.

Taking the observation above into account, it suffices to notice that the CNOT gate is found at the third level of $C(B_S^2, E_S^n)$, and the result follows by Proposition 6. □

It is worth mentioning that, from a physical point of view, the hierarchy $C(B_S^n, E_S^n)$ that we considered above would match the hierarchy of gates that could be implemented using generalized gate teleportation with Basis S and assuming that the Hadamard gate and the gates in Basis S can be applied fault tolerantly on individual qubits.

5.4 | Basis R

The family

$$B_R := \frac{1}{2}\left\{\begin{pmatrix} 1+i & 1-i \\ 1+i & -1+i \end{pmatrix}, \begin{pmatrix} 1-i & 1+i \\ 1-i & -1-i \end{pmatrix}, \right.$$
$$\left.\begin{pmatrix} 1-i & 1+i \\ -1+i & 1+i \end{pmatrix}, \begin{pmatrix} 1+i & 1-i \\ -1-i & 1-i \end{pmatrix}\right\}$$

forms an orthogonal unitary basis. Its projection onto $[U(2)]$ does not form a group, and the group that it generates is isomorphic to the alternating group of degree 4, which has 12 elements. It then follows that $C(B_R) \neq C_*(B_R)$. According to our computations, both of these hierarchies have a T-level greater than 16.

Because of the nature of the symbolic computations involved, the hierarchies generated by Basis R were much more difficult to compute. However, we were able to determine that $C_*(B_R)_2 =$

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$N_{[U(N)]}(B_R)$ has 12 elements and that the second level of $C(B_R)$ has 24.

These are all examples of orthogonal unitary bases for one-qubit systems, which—as we have illustrated—can generate orthogonal unitary bases for qubit systems of any dimension through tensor products. In addition, in systems with more than a prime-dimensional qudit, it is possible to consider nice error bases beyond the Pauli basis [10, 11].

6 | Conclusions and Future Work

In this work, we have discussed how the quantum gate teleportation mechanism can be extended beyond the Pauli group, showing how it can be applied on any orthogonal unitary basis of operators and how each of these bases induces its own Clifford hierarchy. In pursuit of a clearer intuition on these hierarchies, we have also introduced the notion of quasinormalizers, which generalize the construction that determines which sets of gates can be implemented using gate teleportation mechanisms. From these definitions, we have been able to deduce general properties that we then particularized and applied to the study of Clifford hierarchies.

We have been able to prove how no bounded number of applications of teleportation mechanisms can implement an infinite amount of fault-tolerant gates out of a finite collection of fault-tolerant gates. This was a known fact for the usual Clifford hierarchy, but we have generalized it—with a simple and very direct proof—to the hierarchy induced by any general gate teleportation protocol. We have also deduced general properties about the structure and cardinality of the levels of the Clifford hierarchies.

Leveraging our theoretical framework, we have introduced new Clifford hierarchies, studied their properties, and computed their first few levels. Based on the results in Section 5, we see potential for alternative hierarchies to make the fault-tolerant implementation of some quantum gates more efficient, which is something that still has to be investigated. In addition to this, we have introduced some theoretical results that connect Clifford hierarchies with the exact synthesis of quantum gates whose coordinate matrices have cyclotomic integer entries.

There are several possible areas for future work. One of them is extending our theoretical study of quasinormalizers, looking for properties that could be applied to the study of Clifford hierarchies. Another interesting area of work would be the study of a full characterization of the Basis S hierarchy and of all the hierarchies that may share a similar composition.

Author Contributions

Samuel González-Castillo: conceptualization, investigation, writing – original draft, methodology, validation, formal analysis, data curation. **Elías F. Combarro:** conceptualization, investigation, funding acquisition, methodology, validation, writing – review and editing, formal analysis, supervision. **Ignacio F. Rúa:** conceptualization, investigation, funding acquisition, methodology, validation, writing – review and editing, formal analysis, supervision. **José Ranilla:** conceptualization, investigation, funding acquisition, methodology, validation, writing – review and editing, formal analysis, data curation.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study

Endnotes

[1] In rigor, in the expression below, $C(B,S)_k$ should be any set of unitaries E, such that $[E] = C(B,S)_k$.

[2] This result could be trivially generalized to subsets of any general linear group over the complex numbers. However, we have chosen not to do it, for it would add nothing to our discussion.

[3] Keep in mind that, given any operator V with $p(V)(x) = (x-\lambda_1)\cdots(x-\lambda_r)$, we must trivially have $p(\alpha V)(x) = (x-\alpha\lambda_1)\cdots(x-\alpha\lambda_r)$ for any scalar α.

[4] In the article [8], *conjugate tuples* are defined as tuples of elements that are conjugate to some fixed generators of the Pauli group. These tuples are k-closed.

[5] We mean that, for any natural k, $C(\sigma(B),S)_k = \sigma(C(B,\sigma^{-1}(S))_k)$.

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Appendix A

An Algorithm for the Computation of General Clifford Hierarchies

de Silva introduced an algorithm for enumerating the elements of any level of the usual Clifford hierarchy [8, §3] (generated by the Pauli group). Building upon the ideas present in that work, we will now discuss an algorithm that will enable us to compute the elements of any level of any general Clifford hierarchy $C(B,S)$, only under the assumption that S is finite. We have used this algorithm to symbolically compute the first few levels of the one-qubit hierarchies that we discussed in Section 5.

Before being able to introduce the algorithm that will allow us to enumerate all the elements of general Clifford hierarchies, we need to define a few preliminary notions.

Given a natural N, let $(V_1, \dots, V_n) \subseteq U(N)$ be representatives of all the elements of a finite family $B \subseteq [U(N)]$, and let $(G_1, \dots, G_k)$ be representatives of elements of B. A *blueprint* ξ of B for $(G_1, \dots, G_k)$ is a list $\{\{\xi_{rs}\}_{s=1}^{u_r}\}_{r=1}^{n}$ of n finite sequences of natural numbers such that, for every $r = 1, \dots, n$,

$$V_r = G_{\xi_{r1}} \dots G_{\xi_{ru_r}}.$$

Under these same hypotheses, a *multiplication table* M for B is an $n \times n \times n$ tensor such that

$$V_r \cdot V_s = \sum_{k=1}^{n} M_{rsk} V_k.$$

for every $r, s = 1, \dots, n$.

Given any sequence of unitary operators $H := (H_1, \dots, H_k)$, the result of *applying* the blueprint ξ on H is the sequence $(W_1, \dots, W_n)$ where

$$W_r = H_{\xi_{r1}} \dots H_{\xi_{ru_r}},$$

for $r = 1, \dots n$.

These are all the additional tools that we will need for now. Next, let us discuss the components of our enumeration algorithm.

Assume that we have a certain level of a Clifford hierarchy and we are trying to compute the next one. These levels are subsets of $[U(N)]$, so we will be storing them as sets of representatives in $U(N)$. This raises the question of how we can use the representatives of the levels to which we have access in order to find those of the levels above. The following result addresses this question.

Lemma A1. *Let N be a natural number and consider two subsets $B, D \subseteq U(N)$, and a $Q \in U(N)$. We have $c_{[Q]}([B]) = [D]$ if and only if there exists a subset $E \subseteq U(N)$ such that $[E] = [D]$ and $c_Q(B) = E$. In other words, $c_{[Q]}([B]) = [D]$ if and only if $[c_Q(B)] = [D]$.*

Consequently, there is a bijective correspondence between $N_{[U(N)]}([B],[S])$ and the sets of representatives of elements of S which are conjugate to B. Translating this into our context, this means that any $[Q]$ belongs to the next level of the hierarchy if and only if there exists a representative E in the current level and some phase $\alpha \in U(1)$, such that $c_Q(B) = \alpha D$. Now, we need to find a way of determining, out of our sets of representatives, those conjugates with their appropriate phases. That is what the following result is for.

Lemma A2. *Given a natural number N, let $B \subseteq U(N)$ be a set with a blueprint ξ for generators $(G_1, \dots, G_k)$ and assume that there exists a natural d_r for every $r = 1, \dots, k$ such that $(G_r)^{d_r} = \omega_r I$ for some phase $\omega_r \in U(1)$. Consider a pair of sets $E, D \subseteq U(N)$ such that $[D] = [E]$. If B is conjugate to D, then there must exist some $H_1, \dots, H_k \in E$ such that, for some $\alpha_1, \dots, \alpha_k \in U(1)$, D is obtained upon applying the blueprint ξ on $(\alpha_1 H_1, \dots, \alpha_k H_k)$. In addition to this, for every $r = 1, \dots, k$, we must have $(H_r)^{d_r} = \theta_r I$, with $\theta_r \in U(1)$ and*

$$\alpha_r = \exp\left(\frac{2\pi i}{d_r} t_r\right) \cdot \left(\frac{\theta_r}{\omega_r}\right)^{1/d_r},$$

for some $t_r \in \{1, \dots, d_r\}$.

Proof. By virtue of Lemma A1, let $Q \in U(N)$ be the operator such that $c_Q(B) = E$. Because conjugation defines an automorphism in the algebra of linear operators, it is clear that E must have ξ as a blueprint on the generators $c_Q(G_r)$ $(r = 1, \dots, k)$—and these generators will only differ by the phase factors α_r from some corresponding elements H_r in D.

Let us fix some $r = 1, \dots, k$. Because $[G_r]$ has order d_r, so must $[c_Q(G_r)] = [H_r]$. Moreover, as conjugation preserves eigenvalues, we know that $(G_r)^{d_r} = \omega_r I_r$; hence, $\alpha_r^{d_r} = \omega_r/\theta_r$ and the stated condition on α_r follows. □

In summary, if—for a fixed blueprint—B has some generators $(G_r)_r$ such that the operators $(G_r^{d_r})_r$ are diagonal, in order to look for conjugates of $[B]$ in $[U(N)]$, we may restrict ourselves to the sets obtained by applying the blueprint of B on elements $(H_r)_r$ for which $(H_r^{d_r})_r$ are diagonal and consider the phases $(\alpha_r)_r$ specified in the result. This narrows down our search to a finite set. But we still need to figure out a way of determining when a candidate set is indeed a conjugate and, in case it is, a way of computing its corresponding conjugation operators. The following result takes care of that.

Lemma A3. *Let $B \subseteq U(N)$ be an ordered orthogonal unitary basis of operators, and let $E \subseteq U(N)$ be a sequence with N^2 elements. There exists a $Q \in U(N)$ such that $c_Q(B) = E$ if and only if E is orthogonal and there exists an ordering of E under which it has the same multiplication table as B. If this is the case, for any such ordering, Q is unique up to phase and can be constructed as follows. Any normalized eigenvector of the operator*

$$\sum_{r=1}^{N^2} \frac{1}{\operatorname{tr}(B_r^\dagger B_r)} \cdot \overline{\langle 0|B_r|0\rangle} \cdot E_r$$

for Eigenvalue 1 is equal to $Q|0\rangle$ for a choice of phase of Q. The action of Q on the remaining basis elements would then be given by

$$Q|t\rangle = \sum_{r=1}^{N^2} \frac{1}{\operatorname{tr}(B_r^\dagger B_r)} \cdot \overline{\langle t|B_r|0\rangle} \cdot E_r(Q|0\rangle).$$

Proof. The function that maps the elements of B to the elements of E preserving their order can be extended to an automorphism of the algebra of operators on $\mathbb{C}^N$ if and only if E is an orthogonal basis and has the same multiplication and phase tables as B. According to the Skolem–Noether theorem [19, §14.3], all the automorphisms of this algebra are inner; hence, there exists a $Q \in U(N)$ such that $c_Q(B) = E$ if and only if E is an orthogonal (unitary) basis, and for some ordering, it has the same multiplication and phase tables as B. The uniqueness of Q up to phase for any suitable ordering of E follows from the fact that two unitary operators induce the same conjugation action if and only if they only differ by a global phase factor.

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Assume the existence of such a Q, and let us compute the possible images of Q under the computational basis vectors. It is obvious that the vector $Q|0\rangle$ will be an eigenvector of the operator

$$Q|0\rangle\langle 0|Q^{-1} = c_Q(|0\rangle\langle,0|)$$

with Eigenvalue 1. In addition, the eigenspace associated to 1 is one dimensional, so any unit eigenvector of $c_Q(|0\rangle\langle 0|)$ will be the image of $Q|0\rangle$ up to some phase (which, at this stage, we can safely ignore). Moreover, recalling that we are considering orthogonality with respect to the Hilbert–Schmidt inner product, we can notice that

$$\begin{aligned} c_Q(|0\rangle\langle 0|) &= c_Q\left(\sum_r \frac{1}{\mathrm{tr}(B_r^\dagger B_r)}\mathrm{tr}\left(B_r^\dagger|0\rangle\langle 0|\right)\cdot B_r\right) \\ &= \sum_r \frac{1}{\mathrm{tr}(B_r^\dagger B_r)}\langle 0|B_r^\dagger|0\rangle\cdot c_Q(B_r) \\ &= \sum_r \frac{1}{\mathrm{tr}(B_r^\dagger B_r)}\overline{\langle 0|B_r|0\rangle}E_r. \end{aligned}$$

Once a value for $Q|0\rangle$ has been fixed—with any phase—we can easily compute the images of the remaining vectors of the computational basis as

$$\begin{aligned} Q|t\rangle &= \left(Q|t\rangle\langle 0|Q^{-1}\right)Q|0\rangle = c_Q(|0\rangle\langle 0|)Q|0\rangle \\ &= c_Q\left(\sum_r \frac{1}{\mathrm{tr}(B_r^\dagger B_r)}\mathrm{tr}\left(B_r^\dagger|t\rangle\langle 0|\right)\cdot B_r\right)Q|0\rangle \\ &= \sum_r \frac{1}{\mathrm{tr}(B_r^\dagger B_r)}\langle 0|B_r^\dagger|t\rangle\cdot c_Q(B_r)Q|0\rangle \\ &= \sum_r \frac{1}{\mathrm{tr}(B_r^\dagger B_r)}\overline{\langle t|B_r|0\rangle}\cdot E_r(Q|0\rangle). \end{aligned}$$

This concludes the proof. □

We now have all the pieces gathered to construct the algorithm.

Algorithm A1. (Clifford hierarchy enumeration). Let B be an ordered orthogonal unitary basis of operators on $\mathbb{C}^N$ for which $[B]$ is a group, and let $S \subseteq U(N)$ be a subset including B. Consider a blueprint ξ of B with some generators $(G_1, \dots, G_k)$, and let $(d_1, \dots, d_k)$ be the smallest natural numbers such that, for every r, the operators $G_r^{d_r}$ are equal to $\theta_r I_N$, for some $\theta_r \in U(1)$.

Assume that C is a set of representatives of all the elements of some level of the Clifford hierarchy $C(B, S)$.

Combining Lemmas A2 and A3, we know that the elements of the next level C' are in perfect correspondence with all the k-tuples $(\alpha_1 H_1, \dots, \alpha_k H_k)$, where $H_r \in C$ and $\alpha_r \in U(1)$ such that:

- For every $r = 1, \dots, k$, the operator $(H_r)^{d_r}$ is diagonal.
- The phases α_r are obtained as in Lemma A2.
- The ordered set generated by applying the blueprint ξ on H is an orthogonal basis and has the same multiplication and phase tables as B.

Taking this into account, we can compute a set of representatives for the next level of the hierarchy as follows:

1. Find the collection R of all k-tuples $(H_1, \dots, H_k)$ with $H_r \in C$ such that, for every index r, the natural d_r is the smallest number such that $(H_r)^{d_r}$ is a diagonal operator.
2. For each tuple H in R, consider the result of applying the blueprint ξ on H. If the resulting ordered set is not an orthogonal basis or is not a subset of S, remove it from R.
3. Create an empty list L. For each tuple $H = (H_1, \dots, H_k)$ in R:
   a. Assume that, for every $r = 1, \dots, k$, $(H_r)^{d_r} = \omega_r I_n$. Consider the collection of the $d_1 \cdots d_r$ k-tuples of the form $(\alpha_r H_r)_r$, where, for every r, α_r may take the values $\exp(2\pi i t_r/d_r)(\theta_r/\omega_r)^{1/d_r}$ for $t_r = 1, \dots, d_r$.
   b. For each tuple in the collection above, apply the blueprint ξ on it. If the resulting ordered set E has the same multiplication table as B, add E to the list L.
4. For every ordered set in the list L, construct an operator Q, as in Lemma A3, such that $c_Q(B) = E$. The set of representatives of the next level of the hierarchy is the set of all of these operators.

We can deduce a simple yet interesting result from this algorithm.

Corollary A1. *Consider an N-dimensional quantum system with an orthogonal unitary basis B and a subset $S \subseteq U(N)$. If all the entries of the coordinate matrices of B and S are algebraic numbers, then every gate in the $C(B, S)$ hierarchy has a representative with a coordinate matrix in which all entries are algebraic numbers.*

The generalization of this algorithm to arbitrary Clifford hierarchies was conceptually straightforward; in this regard, our main contributions are Lemmas A2 and A3 and the incorporation of blueprints and multiplication tables into the algorithm.

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